IT PROJECT MANAGEMENT · TEACHER REVIEW PACKAGE · VERSION 1.0

ClassroomCast — Project Scope & Summary for Teacher Review

Local Network Screen Sharing and Access Control System

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Meeting	10 November 2026
Status	Planning and analysis documents only. NOT IMPLEMENTED NOT TESTED
Sources	Files 01–08 in `classroomCast_v2/docs/submission-2026-11-10` , checked on 9 November 2026

Status labels used in this document: CONFIRMED stated in the source files · PROPOSED team recommendation, not approved · PENDING needs a teacher, team or campus IT decision · CONFLICTING the source files disagree · NOT IMPLEMENTED · NOT TESTED

1. Project overview

Problem. In a classroom of 50–150 students, many students cannot read the projector screen clearly. Teachers also have no simple way to stop students from opening AI assistants, coding agents or other distracting websites during labs and exams. Today, blocking websites means changing router settings by hand for every class.

Vision PROPOSED. Teachers can show their screen directly on student laptops over the local network, manage who joins, and apply approved website restrictions from one dashboard. The system keeps working without an internet connection.

Three main functions

1. **Classroom broadcasting.** The teacher shares a screen, window or tab with supported audio. Students watch in a web browser and install no application.
2. **Classroom management.** The teacher creates a class with a join code and QR code, then approves, rejects, removes or blocks participants.
3. **Software-managed website restrictions.** Authorized administrators choose profiles such as *Normal Class*, *Programming Lab* and *Exam Mode*. A local DNS filter enforces the rules. The team proposes **AdGuard Home** for this: a self-hosted service that refuses to look up blocked website names.

Key figures (verified against Files 01–08)

63	4	68	15	18	90
original requirements	user roles	feature entries in 6 areas	use cases (UC01–UC15)	process flows (F01–F18)	test specifications (63 + 27)

Requirement priorities as given in the specification: **31 Must, 7 Should, 3 Could**. The 22 campus-network (CN) and teacher-owned-router (OR) obligations have no priority label. The team proposes treating them as Must PENDING.

2. Objectives and project scope

Objectives PROPOSED. Each objective has a measurable check in File 01.

- Deliver teacher screen and audio to admitted students on the local network (F04–F07).
- Allow only the owning teacher to control a class and its participants (role and admission checks).
- Keep working when the internet (WAN) is disconnected (test AT07).
- Apply only authorized website restrictions, and verify them from a student device (F10).
- Restore the previous network policy safely when a class ends, a rule expires, or an emergency occurs (AT05).
- Report capacity and quality honestly, keeping physical-device and simulated results separate (AT02/AT03).

Target users. One teacher and 50–150 students per classroom CONFIRMED, plus system and network administrators.

In scope CONFIRMED

- Local screen and audio broadcasting through a shared media server
- Class creation, join code, QR code and admission control
- Policy profiles, categories, domains and exceptions
- Automatic apply when a class starts and restore when it ends
- Enforcement status, audit log, emergency rollback, rule expiry
- Offline operation, diagnostics, and both deployment modes

Out of scope PROPOSED (W01–W05)

- W01 Building our own media server; we reuse an existing one
- W02 A required student app; access stays browser-only
- W03 Recording or archiving class video
- W04 LMS or ERP integration
- W05 Remote control or capture of student screens

Deployment assumptions

Mode	Assumption	Status
A. Campus Wi-Fi / LAN	Uses the existing campus network. Hosting and DNS changes require written approval from campus IT (CN05). The system must never bypass campus controls.	PENDING
B. Teacher-owned router	A portable network controlled by the team: server plus Wi-Fi access points, fully offline (OR01–OR12).	CONFIRMED in scope
Host	An existing school Ubuntu/Linux server was mentioned. Its hardware and permissions have not been inventoried.	PENDING

Technical constraints

- Browsers allow screen and microphone capture only on secure (HTTPS) pages, so the local server needs a trusted certificate.
- Supported audio sources differ by browser and operating system; system audio is not available everywhere.
- DNS filtering can be bypassed with VPNs, alternate or encrypted DNS, direct IP addresses, or locally installed AI tools. We will document this limitation, not hide it.
- Restrictions must never block ClassroomCast itself (NAC18).
- Router credentials and API secrets stay on the server (NAC17). We collect as little browsing history as possible (NAC19).

3. System roles and product structure

Four roles CONFIRMED (File 03). Permissions are checked on the server. Each role is limited to its own classes or assigned network scope.

Role	Main responsibility	Cannot do
System Administrator (SYS)	Manage accounts, roles, application settings, policy profiles; read audit and health records	Teach, view class media, or change network infrastructure
Teacher (TCH)	Create and run own classes; share screen and audio; admit, remove or block students; select an approved profile	Edit DNS or router settings, see secrets, or edit policy rules
Student (STU)	Request admission; watch and listen; use permitted materials	Publish media, control the class, or change policies
IT / Network Administrator (NET)	Connect DNS, gateway or router integrations; apply, verify and recover network policies	Act without campus IT or equipment-owner authorization

Six capability areas and 68 feature entries CONFIRMED (File 04). The four product pillars in the specification map to these six areas. Area 06 supports all four.

Area	Pillar	Entries	Main requirement IDs
01 Classroom broadcasting	Broadcasting	7	T02–T04, T07, S03–S05
02 Management and participation	Classroom management	11	T01, T05, T06, T08, T10, T11, S01, S02, S08–S10
03 Network access control	Network access control	13	NAC01–09, NAC14, NAC15, NAC18, NAC20
04 Deployment and infrastructure	Network administration	22	CN01–CN10, OR01–OR12
05 Diagnostics, verification, recovery	Network administration	7	S06, S07, T09, NAC10, NAC12, NAC13 + 1 basic-health entry
06 Administration and security	Supports all pillars	8	NAC11, NAC16, NAC17, NAC19 + 4 administration entries

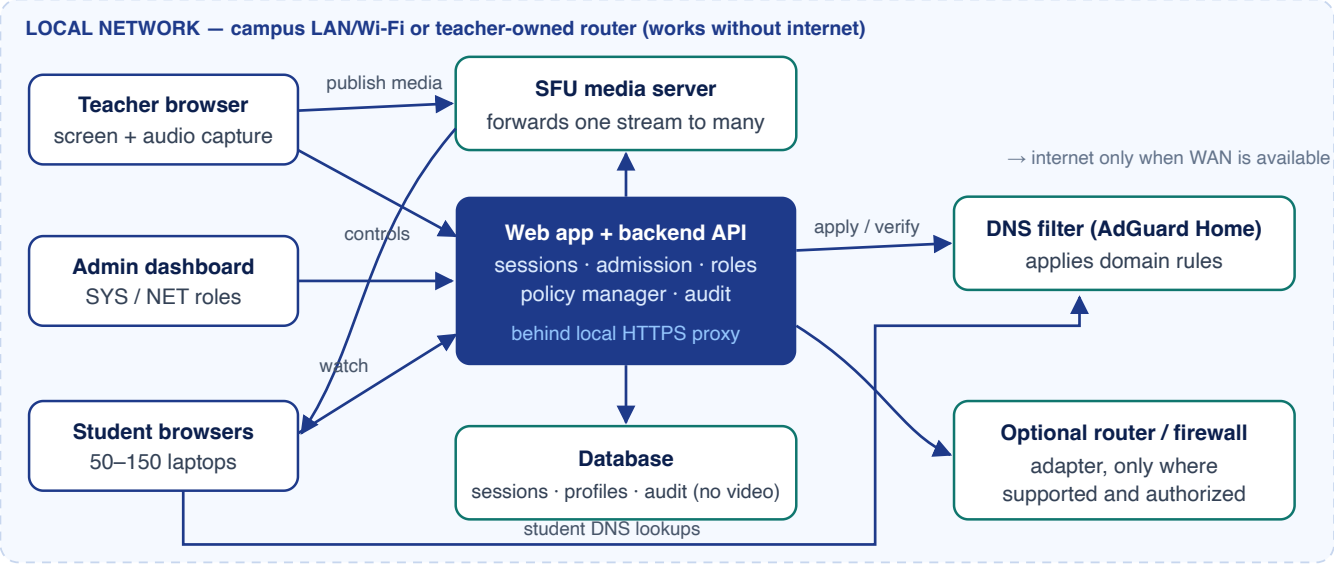
68 entries = 63 original requirements + 2 required obligations with no ID + 3 proposed administration entries (accounts, role assignment, settings). **These entries describe requirements. They are not finished features.** NOT IMPLEMENTED

Traceability CONFIRMED. Every one of the 63 requirements links to a feature entry, a process flow, a use case, a user story and a test case.

Requirement group	Features	Flows	Use cases
NAC01–NAC20 Network access control	Areas 03, 05, 06	F01, F03, F09–F15, F17, F18	UC05, UC06, UC08–UC10, UC12, UC13, UC15
T01–T11 Teacher	Areas 01, 02, 05	F04–F08, F15, F16, F18	UC01–UC04, UC10, UC11, UC13
S01–S10 Student	Areas 01, 02, 05	F05–F07, F12, F15, F16, F18	UC02, UC03, UC07, UC10, UC11, UC13
CN01–CN10 Campus network	Area 04	F02	UC14
OR01–OR12 Teacher-owned router	Area 04	F03	UC15

4. Proposed system architecture

The architecture keeps every component on the local network. One teacher stream goes to a **Selective Forwarding Unit (SFU)**: a media server that receives the stream once and forwards a copy to each admitted student. The teacher's laptop does not have to upload 150 separate streams.



High-level architecture. Box roles CONFIRMED; product names inside boxes PROPOSED.

Design decisions in the specification CONFIRMED

- Use an SFU, not a direct teacher-to-every-student connection.
- Keep the web app, signaling, SFU, policy manager and DNS filter on the local network.
- Keep secrets on the server only. Show the configured policy and the actually enforced policy separately.
- Provide health checks, logs and recovery procedures. Serve the app over local HTTPS for screen capture.

Proposed technology PROPOSED. Final choice after feasibility tests (decision D19).

Layer	Proposed choice
Frontend	React + Vite + TypeScript
Backend API and signaling	Node.js (NestJS)
Media server (SFU)	Self-hosted LiveKit (mediasoup is the alternative named in the specification)
Database	SQLite first; PostgreSQL if needed
DNS enforcement	AdGuard Home
HTTPS proxy / deployment	Caddy · Docker Compose

Data and API design PROPOSED. The conceptual model has 8 entities: User, ClassroomSession, Participant, PolicyProfile/PolicyVersion, EnforcementScope, PolicyLease, PolicyOperation and AuditEvent. A list of 10 API endpoints is drafted, such as `POST /api/sessions` and `POST /api/scopes/:id/restore`. Detailed field and request formats are still to be designed.

5. Proposed MVP scope

Why an MVP? Five members at about 10 hours a week each (File 01) give roughly **320 hours** across the three build sprints (30 Nov – 13 Jan). This figure is **our own calculation, not stated in the source files**. The whole project estimate is 480 hours (420 work + 42 contingency + 18 reserve). This covers design, testing and closing as well as the sprints. The sprints include the year-end holidays. We therefore propose building and demonstrating the core product first in **teacher-owned router mode**, because the team controls that equipment and does not need to wait for campus IT. PROPOSED PENDING teacher approval.

Level A — Build and demonstrate

What will work in the demo	Requirement IDs
Create a class; students join by URL, QR or code with a name; teacher approves	T01, T05, T06, T08, S01, S02, S07, S08
Teacher shares screen and audio; start, stop, pause and resume; quality settings	T02, T03, T04, T07
Students watch live, control volume, use fullscreen, and reconnect after short drops	S03, S04, S05, S06
Profiles, categories, domains, allow or block by default, exceptions	NAC01–NAC04, NAC07, NAC08
Rules apply when a class starts and restore when it ends; status shown; actions logged	NAC05, NAC06, NAC10, NAC11
Emergency rollback, automatic rule expiry, admin-only control, server-side secrets, ClassroomCast never blocked, minimal browsing data, offline rule storage	NAC12, NAC13, NAC16–NAC20
Teacher-owned router setup that works fully offline	OR02, OR03, OR05, OR07, OR11, OR12

All 31 Must requirements are in Level A. Two Should items move up because the core needs them: **S08** (student name, needed for the approval step) and **NAC20** (offline rules, built into AdGuard Home).

Level B — Design and document

Deliverable	Requirement IDs
Campus-mode architecture, setup guide and campus IT approval checklist. We deploy in campus mode only if campus IT gives written approval.	CN01–CN10
Capacity evaluation: test with about 10–20 real laptops, simulate the rest at 50/100/150 viewers, and report the two separately	CN06, OR01, OR04, OR08
Hardware list and portable-kit setup guide	OR06, OR09, OR10

*Moving campus deployment to documentation only is a **scope change**. Under our change-control rule, the teacher must approve it.*

Level C — Optional, only if time allows

Feature	Requirement IDs	Source priority
Streaming diagnostics · file sharing · downloads	T09, T10, S09	Should
Scheduled restrictions · policy import/export	NAC09, NAC14	Should
Session history · hand-raise/questions/polls · group/device policies	T11, S10, NAC15	Could

6. Project schedule and team plan

Timeline CONFIRMED from the course website plan (File 07). Sprint goals PROPOSED.

Phase	Dates	Goal
Preparation	10–17 Nov 2026	Requirements, stories, scope and proposal review; **teacher approval**
Design	18–27 Nov	Architecture, data model, UI, API, test plan; feasibility tests SP01–SP03 (HTTPS, SFU, DNS apply/restore)
Sprint 1	30 Nov – 11 Dec	Create a class, join by code or QR, approve students. One teacher shares a screen with about 3 laptops on our router. AdGuard blocks one test domain.
Sprint 2	14 – 25 Dec	Audio, quality, pause/resume, remove/block, reconnect; profiles and categories; apply at class start and restore at class end
Sprint 3	28 Dec – 13 Jan 2027	Emergency rollback, expiry, audit log, protect local traffic; test with 10–20 laptops; **bug fixing only, no new features**
Testing and release	14 – 27 Jan	Acceptance tests and defect fixes; demo and user guide 14–20 Jan; site release 27 Jan
Closing	28 Jan – 3 Feb	Final report, slides and rehearsal; **final presentation 3 Feb**

The sprint goals above follow File 07's sprint focus. They add the teacher-owned router as the first environment. CONFLICTING File 01 and the Scrum plan in File 07 still describe an older **six-sprint** plan (2 Nov – 24 Jan). File 07 says the three-sprint plan replaces it. The teacher should confirm.

Team roles PROPOSED. HENG NGUON HOUR is Product Owner and HONG DYHENGMESA is Scrum Master; all five members are Developers.

Task owners CONFLICTING. The sources contain three different allocations. The edited planning document is shown below, followed by the differences in the newer timeline.

Member	Edited planning document	Newer timeline differs
HENG NGUON HOUR	Requirements/stories, sign-in/roles, search/reports/polish, presentation	—
CHANTHA MENGKONG	Architecture/data model, main-data operations, acceptance/bug fixes	Core use cases, demo deployment
BUN SENGLEANG	UI mockups, core use cases, demo deployment	Main data, acceptance
HAK SENGKEA	Test plan, unit/integration tests, user guide	—
HONG DYHENGMESA	Sprint planning/backlog, remaining use cases, final report	Also shares UI work

A third allocation, the backlog leads for CC-001 to CC-026, follows the old six-sprint plan. Task names such as "search/reports/polish" come from a generic template and should be renamed to match ClassroomCast features.

Key dependencies and schedule risks

Risk / dependency	Response
Campus IT may not authorize DNS changes (R01)	Develop on the teacher-owned router; campus mode stays Level B until written approval
Wi-Fi or server capacity unknown (R02)	Wired server, staged tests, equipment inventory before Sprint 1
Trusted HTTPS on student laptops (R03)	Two-laptop HTTPS test during the design phase
Policy restore fails after a crash (R04)	Store the previous policy first; restart and restore drills in Sprint 3
Work exceeds available hours (R07)	MVP levels A/B/C; defer Level C first
Holidays inside Sprints 2–3	Confirm each member's availability; no new features in Sprint 3

7. Testing and evaluation

Current status: test cases are specified, but none has been run. NOT TESTED All five feasibility tests (SP01–SP05) are also *Not run*. No result in this package is a pass.

Test inventory CONFIRMED (File 08)

Test group	Count	What it checks
Requirement test cases (TC-NAC, TC-T, TC-S, TC-CN, TC-OR)	63	One test case for each original requirement
Admission edge cases (TC-ADM01–07)	7	Invalid codes, rejection, last-seat race, unauthorized approval
Architectural cases (TC-AUTH, LOCAL, SFU, RECOVERY, PRIVACY)	5	Permissions, offline operation, SFU path, recovery, secret protection
Isolated failure scenarios (TC-FAIL01–08)	8	Internet loss, Wi-Fi loss, backend/SFU/DNS/HTTPS failure
Workflow race cases (TC-CTRL01–04, TC-CLOSE01–03)	7	Late or repeated actions during stop and end class
Total	**90**	

Test levels

- **Functional:** class creation, joining, broadcasting, roster control, profiles and full teacher/student workflows.
- **Security:** cross-class access is denied, students cannot publish, revoked students cannot rejoin, secrets never reach the browser, and audit records exclude browsing history.
- **Network-policy verification and restoration:** from a student laptop, a blocked domain fails and an allowed domain works. ClassroomCast stays reachable throughout. After a class ends, an emergency or an expiry, the previous policy returns. A saved profile is never shown as “enforced” without this check.
- **Performance:** staged 50 / 100 / 150 viewers for 30 minutes each, measuring delay, freezes, packet loss and server/Wi-Fi load.

Proposed acceptance targets PROPOSED (need agreement, decision D12)

ID	Target
AT01	Join page loads within 5 s on the local network
AT02	720p at 15 fps; screen-to-student delay ≤ 1 s (95th percentile)
AT03	50/100/150 viewers for 30 min with playback active in ≥ 98% of checks
AT04 / AT05	Policy applied and verified within 30 s; previous policy restored within 30 s
AT06	Reconnect within 15 s after a 10 s drop, in at least 9 of 10 attempts
AT07	A fresh browser joins a class with the internet disconnected

Physical versus simulated clients. Real laptops on Wi-Fi test the radio network and the browsers. Simulated viewers test only server load. We will report the two separately and **will not claim 150-student capacity until it is measured**.

8. Teacher approval and next steps

Decisions requested at the meeting PENDING

#	Decision	Our proposal
1	**MVP scope.** Approve Levels A (build), B (document) and C (optional)?	Approve. This includes moving S08 and NAC20 into Level A.
2	**Capacity evaluation.** Is testing with 10–20 real laptops plus simulated viewers acceptable for the 150-student target?	Yes, reported separately (D12)
3	**Campus mode.** Deliver it as design and documentation unless campus IT gives written approval? Who is the campus IT contact?	Yes, as a recorded scope change
4	**Schedule.** Is the 25 Jan editing deadline separate from the 27 Jan release and the 3 Feb presentation? Does the three-sprint plan replace the six-sprint plan?	Finish internal review by 22 Jan
5	**Task owners.** Which of the three allocations is authoritative?	Team agrees one list at this meeting
6	**Acceptance targets.** Accept AT01–AT07 as written, or adjust them?	Accept as starting targets
7	**Exclusions.** Approve the out-of-scope items W01–W05?	Approve
8	**Technology.** Accept the proposed stack, subject to feasibility tests (D19)?	Accept as a proposal

Next steps after approval

1. Record the teacher's decisions in the decision log (`07-teacher-review/03-feedback-and-decision-log.md`).
2. Update the backlog and sprint plan in File 07 to match the approved MVP and one agreed task allocation.
3. List the available equipment: router, access points, server or mini-PC, and test laptops.
4. During the design phase (18–27 Nov), run the feasibility tests: trusted HTTPS (SP01), SFU screen and audio with two viewers (SP02), and AdGuard apply/restore (SP03).
5. Contact campus IT about DNS authorization (CN05). Record the answer, whatever it is.
6. Rename the generic task names to ClassroomCast features before Sprint 1 starts on 30 Nov.

This document is a summary. The full detail is in Files 01–08, and the fact checks are in *ClassroomCast_Source_Verification.md*.